

Aurora Caton

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Education

Lehigh University College of Business, Bethlehem, PA

Completed 05/2026

- B.S. in Business and Economics, Major in Business Analytics
- Minor in Population Health (Data Science Focus)
- Relevant Coursework: Advanced Topics in Data Analytics & Modeling, Managing Information Systems, Bio-Statistics, Data Science Algorithms Lab, Statistics, Database Management, Python Applications, Artificial Intelligence Applications, Finance, Financial & Managerial Accounting, Java Applications

Technical/Research Experience

Machine Learning Research Associate, Lehigh University, Bethlehem, PA

05/2025 - 12/2025

- Developed neural network and ensemble machine learning models in Python to predict Drug-Induced Autoimmunity (DIA), supporting early- stage drug safety assessment and clinical research.
- Built and optimized TensorFlow and Scikit-learn predictive pipelines using feature engineering, hyperparameter tuning, and threshold optimization to improve prediction of DIA-positive compounds.
- Evaluated model performance using ROC-AUC, precision, recall, confusion matrices and cross-validation, achieving ~80% cross-validation accuracy and improving confidence in model reliability for research applications

Business Analytics Case Competition Finalist, Manhattan University

02/2026 - 05/2026

- Built machine learning and Natural Language Processing (NLP) models to classify safety event severity and predict corrective actions from large-scale structured and unstructured datasets..
- Developed an AI-powered decision-support application using a multi-agent predictive analytics framework to assist with risk assessment and evidence-based decision making.

Campus and Community Involvement

Institute for Healthcare Improvement Chapter, Bethlehem, PA, *President*

11/2023 - 05/2026

- Led club initiatives focused on healthcare system improvement and collaboration with the Allentown Health Bureau on opioid-related community projects, strengthening community-health partnerships.

Lehigh Silicon Valley Baker Institute, Bethlehem PA, *Participant*

11/2024 - 01/2025

- Completed a highly selective program immersing students in Silicon Valley's entrepreneurial and technology landscape through direct engagement with industry leaders.
- Participated in collaborative case studies and pitched strategic solutions to technology startups, strengthening problem-solving and presentation skills.

Business Analytics Club, Bethlehem, PA, *Member*

06/2023 - 05/2026

- Built regression and predictive analytics models to identify demographic, behavioral, and physiological factors associated with increased diabetes risk, helping evaluate patterns linked to disease development.
- Applied machine learning techniques, including Decision Trees, Random Forest, and SVMs, to classify mobile phone price ranges using hardware and performance features, supporting more accurate and data-driven pricing strategies.

Skills & Certifications

Technical Skills

- Machine Learning & Data Science: TensorFlow, Scikit-learn, Neural Networks, Random Forest, Logistic Regression, Decision Trees, Ensemble Methods, Predictive Modeling, Feature Engineering, Time Series Analysis, Hypothesis Testing, NLP
- Programming: Python, R, SQL, Java, JavaScript
- Data & Visualization Tools: Pandas, NumPy, Tableau, PowerBI, Excel, Oracle, JupyterLab, Google Colab, RStudio
- Workflow & Collaboration: Git/GitHub, Jira, Scrum, Kanban, Slack, VSCode

Certifications

- Introduction to Python Certification, Intermediate Python Certification, GitHub Certification, LinkedIn Learning, Git from Scratch Course Certification, Institute For Healthcare Improvement Patient Safety Certification